

NORTH SOUTH UNIVERSITY

Department of Electrical and Computer Engineering

EEE 141L / ETE 141L : Electrical Circuit - I Lab



Lab Report Spring 2026

Course Section: 10

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Experiment No. 03

Experiment Name: Kirchhoff's Current Law (KCL), Current Divider Rule (CDR) using Parallel and Ladder Circuit

Date of Performance: 26/02/2026

		GROUP NO. 3			
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Date of Submission: 05/03/2026

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SCORE:	RESUBMIT	REMARKS:	SUBMISSION			
			N	L	SL	R
	UPDATED SCORE		Date:			
	OTHER	PENALTY:	Time:			

□ Experiment Name : Kirchhoff's current Law (KCL), current Divider Rule (CDR) using parallel and ladder circuit.

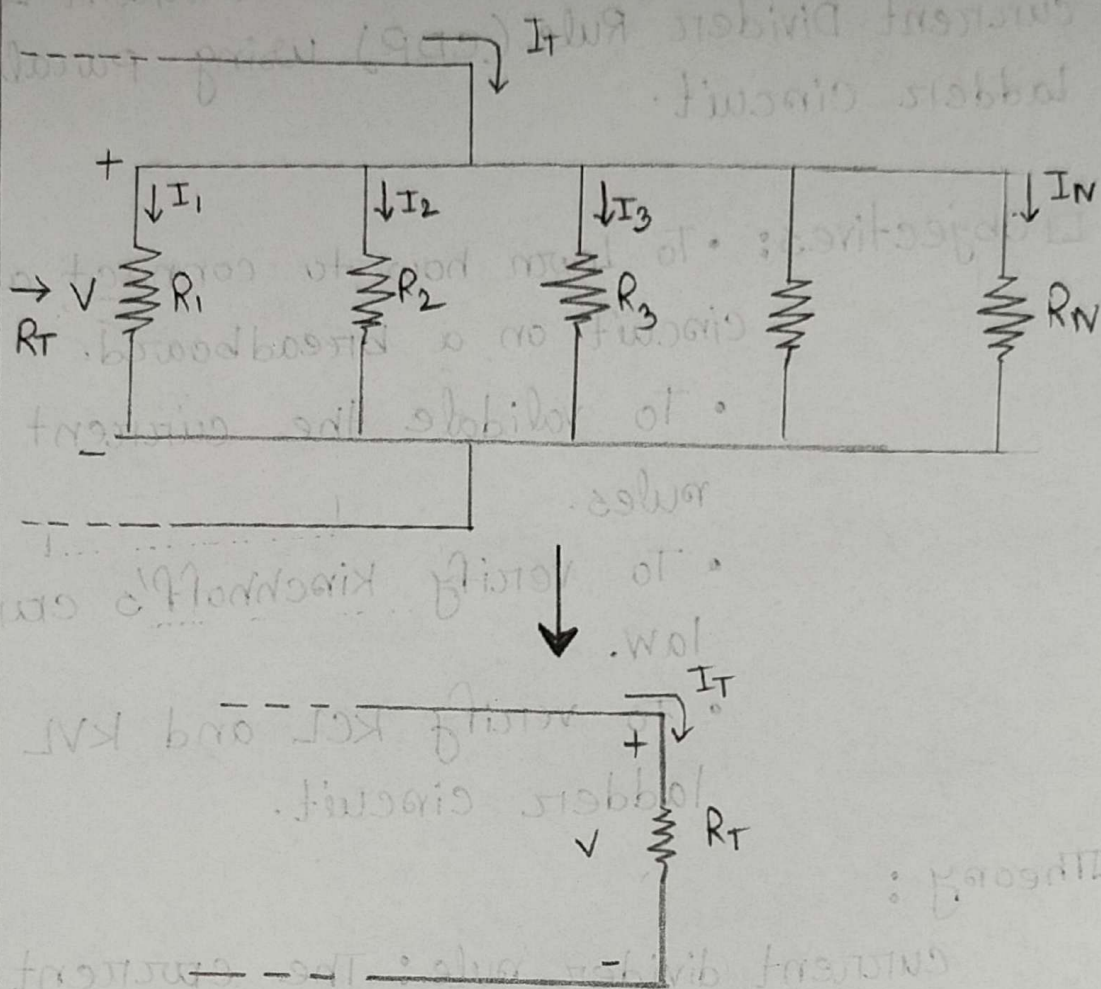
- Objectives:
- To learn how to connect a parallel circuit on a breadboard.
 - To validate the current divider rules.
 - To verify Kirchhoff's current law.
 - To verify KCL and KVL in ladder circuit.

□ Theory :

current divider rule: The current divider rule allows us to calculate the current flowing through each parallel resistive branch as a percentage of the total current.

The main characteristic of parallel circuits is that while they may produce different currents flowing through different branches, the voltage is common to all the connected paths.

$$V_1 = V_2 = V_3 = \dots = V_n$$



The current I_T can then be determined using ohm's law:

$$I_T = \frac{V}{R_T}$$

Since the voltage V is the same across parallel elements, the following is true:

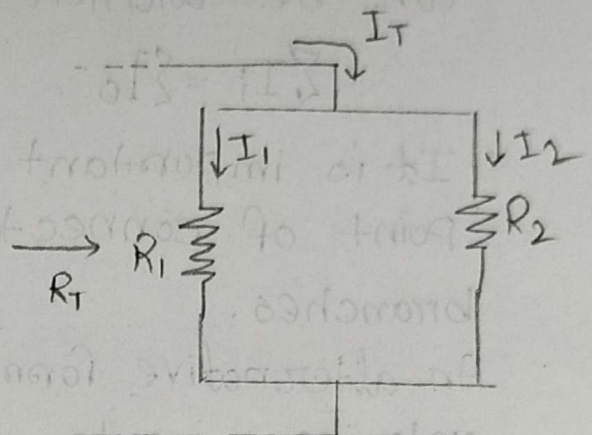
$$V = I_1 R_1 = I_2 R_2 = I_3 R_3 = \dots = I_x R_x$$

where the product $I_x R_x$ refers to any combination in the series.

For the case of two parallel resistors, the total resistance is determined by

$$R_T = \frac{R_1 R_2}{R_1 + R_2}$$

Substituting R_T into the final current divider rule equation results in:



$$I_1 = \left(\frac{R_2}{R_1 + R_2} \right) I_T \quad I_2 = \left(\frac{R_1}{R_1 + R_2} \right) I_T$$

Kirchhoff's current law (KCL): Kirchhoff's current law (KCL) states that the algebraic sum of the currents entering and leaving a node (or junction) of a network is zero.

In other words, the sum of the currents entering a node (or junction) of a network must equal the sum of the currents leaving the same node (or junction).

Mathematically, KCL implies that,

$$\sum_{n=1}^N i_n = 0$$

where, N is the number of branches connected to the node and i_n is the n^{th} current.

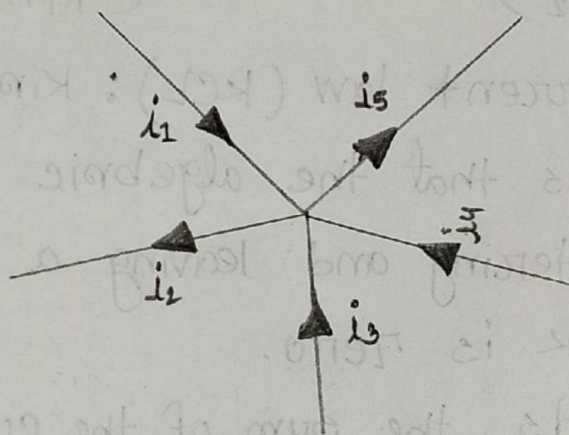
entering the node.

In equation form, Kirchhoff's current law (KCL) can be written as follows:

$$\sum I_i = \sum I_o$$

It is important to know that, a node is the point of connection between two or more branches.

An alternative form of KCL: The sum of the currents entering a node is equal to the sum of the currents leaving the node.



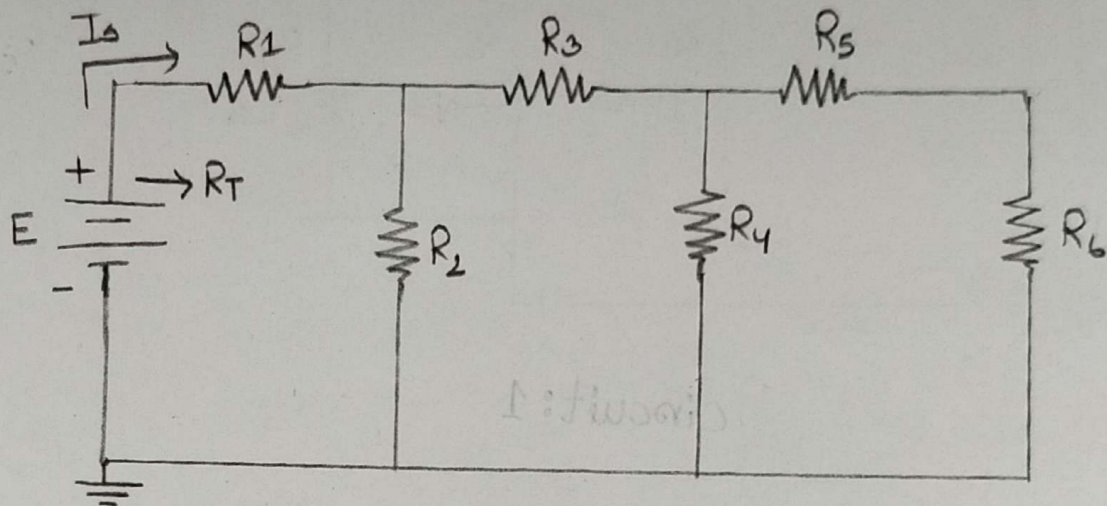
From the above figure we see that, currents i_1 , i_3 , i_4 are entering the node, while currents i_2 and i_5 are leaving it. By applying KCL we get,

$$i_1 - i_2 + i_3 + i_4 = 0$$

rearranging terms gives,

$$i_1 + i_3 + i_4 = i_2 + i_5$$

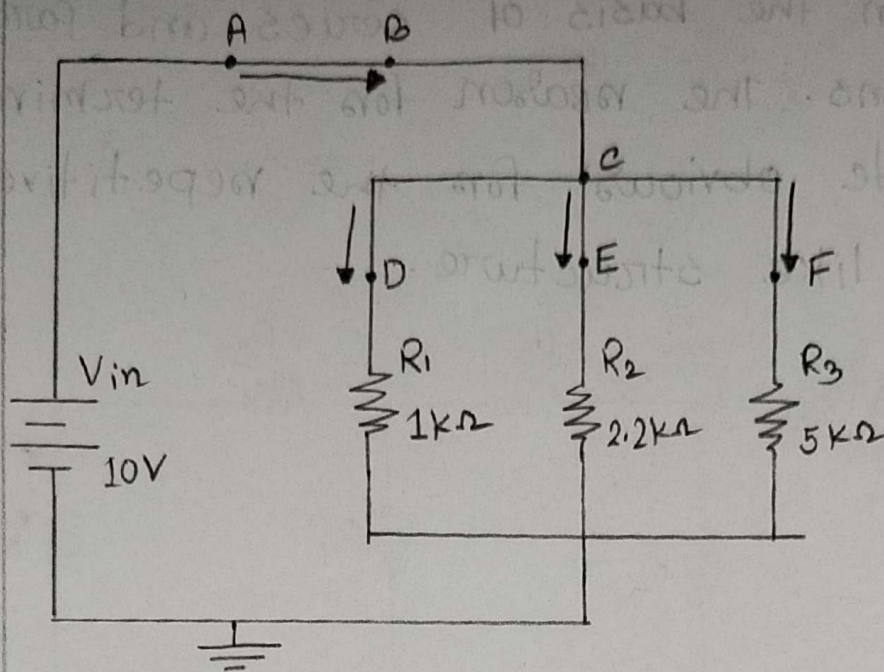
Ladder circuit: The ladder circuit is configured purely on the basis of series and parallel connections. The reason for the terminology is quite obvious for the repetitive ladder like structure.



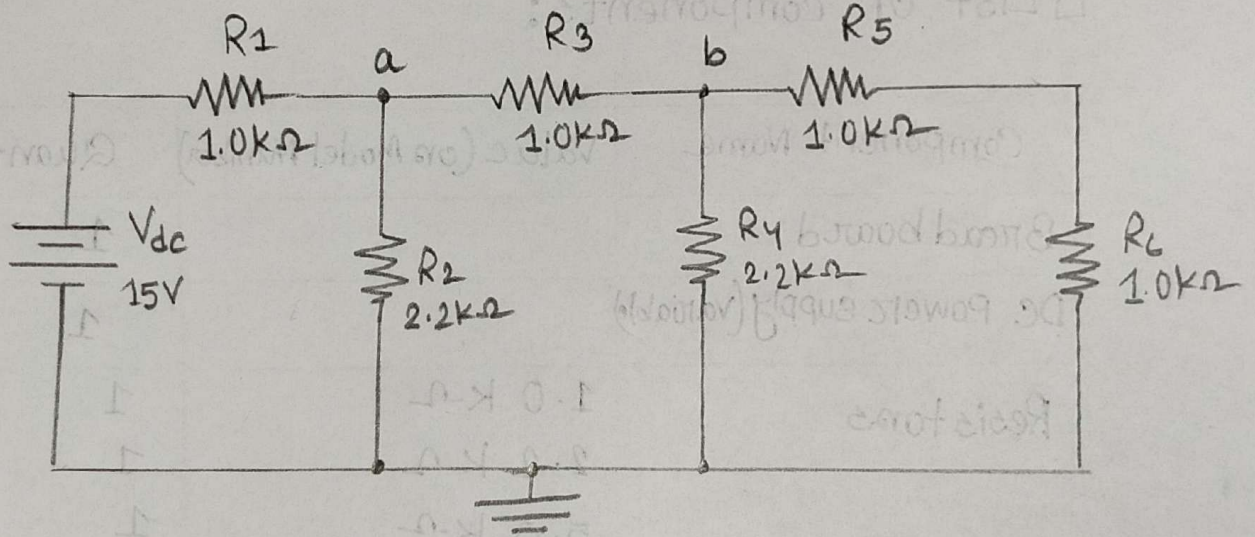
□ List of components:

Component Name	Value (or Model Number)	Quantity
Bread board		1
DC Power supply (variable)		1
Resistors	1.0 k Ω	1
	2.2 k Ω	1
	5.0 k Ω	1
connecting wires		10
Digital multimeter		1

□ Circuit Diagram:



circuit: 1



circuit: 2

LabTech
POWER PROJECT BOARD GIANT
MODEL: LT-375
SparkFun, USA

4V

LabTech
POWER PROJECT BOARD GIANT
MODEL: LT-375
SparkFun, USA

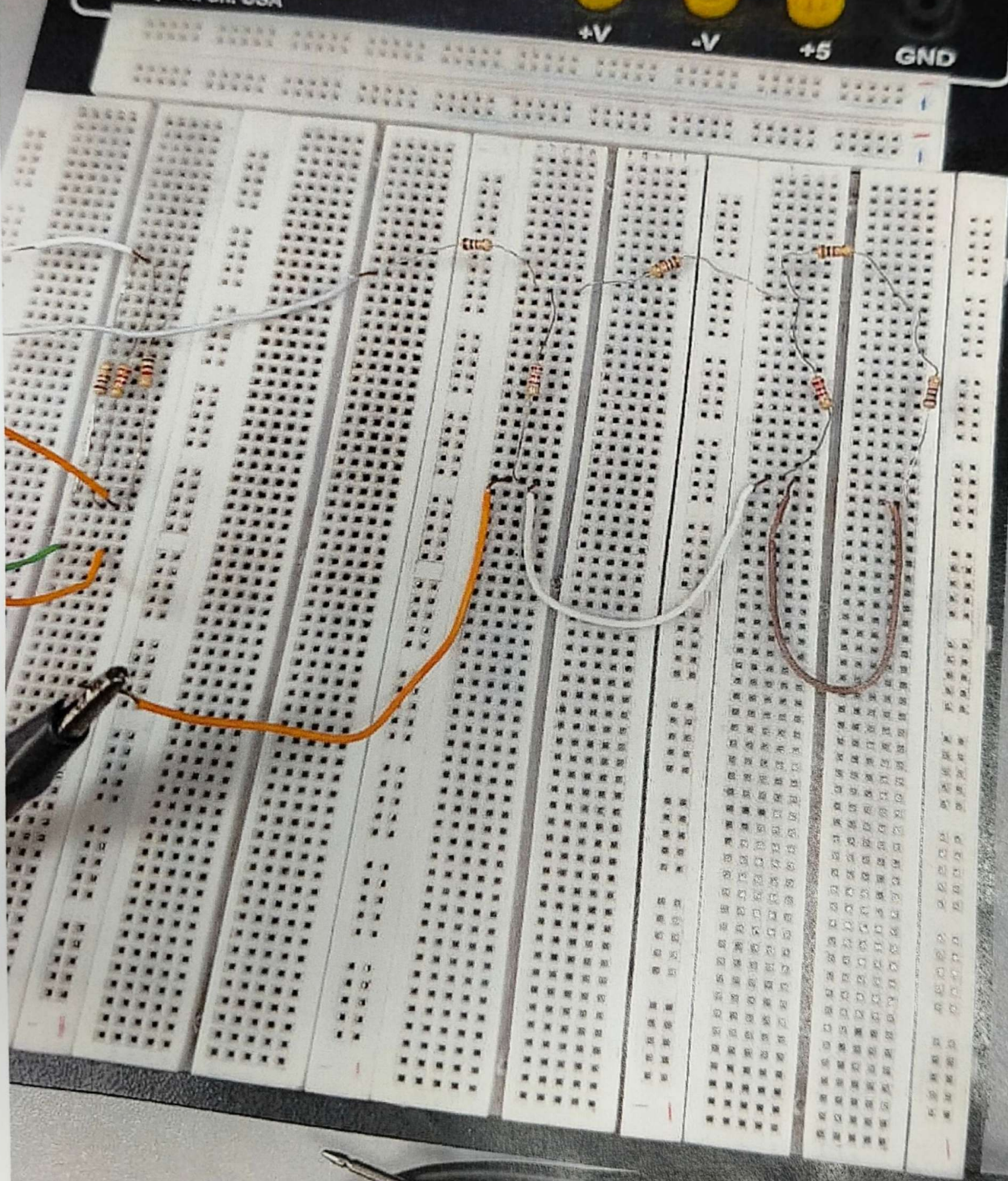


+V

-V

+5

GND



□ Procedure:

1. Identify all the given resistors using color coding and fill in the required columns in Table 1.
2. Measure the resistance of the resistors using the DMM and fill in the required column in Table 1.
3. Calculate the percentage errors of the resistance values.

$$\text{Percentage of Error} = \frac{|\text{Practical value} - \text{Theoretical value}|}{\text{Theoretical value}} \times 100\%$$

4. Build the circuit 1.
5. Connect the DMM (current mode) in series,
 - between A and B to measure I_{in}
 - between C and D to measure I_1
 - between C and E to measure I_2
 - between C and F to measure I_3
6. Record the reading in Table 2. Then fill in table 3.
7. Construct circuit 2.
8. Using a DMM, measure the potential differences across all the resistors in circuit 2. Record all the reading in table 4.
9. Using a DMM, measure the current through All the resistors and record in table 4.

Result / Data:

Table 1:

Theoretical value	Practical value using DMM	% of Error
1 k Ω	0.997 k Ω	0.3%
2.2 k Ω	2.165 k Ω	1.59%
5 k Ω	5.10 k Ω	2.0%
$R_T = R_1 R_2 R_3 = 60.4.3 \Omega$	0.604 k Ω	0.05%

Table 2:

	Experimental reading	Theoretical values	% of Error
I_{in}	16.7 mA	16.545 mA	0.94%
I_{R1}	10.14 mA	10 mA	1.4%
I_{R2}	4.66 mA	4.545 mA	2.53%
I_{R3}	1.97 mA	2 mA	1.5%

Table 3: use data from table 2

Supply current I_{in}	16.7 mA	I_s total current equal to sum of branch current.
Sum of Branch current $I_{R1} + I_{R2} + I_{R3}$	$10.14 + 4.66 + 1.97$ $= 16.77 \text{ mA}$	☐ Equal ☒ Almost Equal ☐ not equal

Table 4: for circuit (2)

Component	Voltage	Current
V_{dc}	15.10V	7.37 mA
R_1	7.35V	7.4 mA
R_2	7.74V	3.53 mA
R_3	3.77V	2.84 mA
R_4	3.95V	1.85 mA
R_5	2.0V	2.83 mA
R_6	2.0V	2.82 mA

□ Question and Answer:

1. State the current division rule (CDR)

Answer: Current division rule states that the total current divided into either of the parallel combination of two resistance or impedance is inversely proportional to the value of Resistance. Can be expressed as

$$I_1 = \frac{R_2}{R_1 + R_2} I_T \quad \text{and} \quad I_2 = \frac{R_1}{R_1 + R_2} I_T$$

Where, I_1 are the current flowing through the first resistor R_1 and I_2 are the current flowing through the second resistor R_2 and I_T is the total current entering the parallel network.

2. State the Kirchhoff's current law (KCL)

Answer: For a parallel path, the total current entering a circuit's junction is exactly equal to the total current leaving the same junction.

KCL can be written as,

$$\sum I_{in} = \sum I_{out}$$

or in algebraic form,

$$\sum I = 0$$

It is important to know that, a node is the point of connection between two or more branches.

3. with the experimental data, verify Kirchhoff's voltage law within each independent closed loop of the circuit.

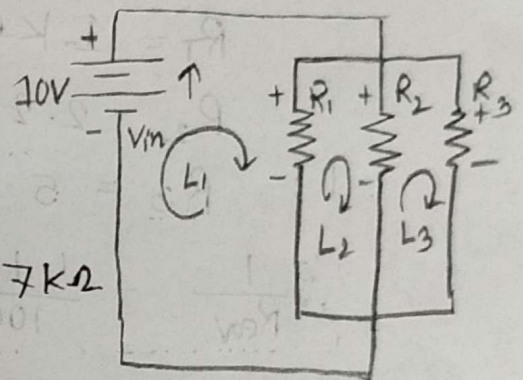
Ans: Loop 1:

$$V_{in} - V_{R_1} = 0$$

$$\Rightarrow V_{in} = I_1 R_1$$

$$\Rightarrow 10V = 10.14 \text{ mA} \times 0.997 \text{ k}\Omega$$

$$\Rightarrow 10V \approx 10.1V$$



Circuit: 1

Loop 2:

$$V_{R_1} - V_{R_2} = 0$$

$$\Rightarrow V_{R_1} = V_{R_2}$$

$$\Rightarrow I_1 R_1 = I_2 R_2$$

$$\Rightarrow 10.14 \text{ mA} \times 0.997 \text{ k}\Omega = 4.66 \text{ mA} \times 2.165 \text{ k}\Omega$$

$$\Rightarrow 10.1V = 10.1V \approx V_{in}$$

Loop 3:

$$V_{R_2} - V_{R_3} = 0$$

$$\Rightarrow V_{R_2} = V_{R_3}$$

$$\Rightarrow I_2 R_2 = I_3 R_3$$

$$\Rightarrow 4.66 \text{ mA} \times 2.165 \text{ k}\Omega = 1.97 \text{ mA} \times 5.1 \text{ k}\Omega$$

$$\Rightarrow 10.1V = 10.04V \approx V_{in}$$

(Ans)

4. showing all steps, calculate the theoretical values in Table 2. Compare theoretical values to your experimental values and explain whether your circuit follows KCL or not.

Ans: given resistances,

$$R_1 = 1 \text{ k}\Omega$$

$$R_2 = 2.2 \text{ k}\Omega$$

$$R_3 = 5 \text{ k}\Omega$$

$$\begin{aligned}\therefore \frac{1}{R_{eq}} &= \frac{1}{1000} + \frac{1}{2200} + \frac{1}{5000} \\ &= 0.001 + 0.0004545 + 0.0002 \\ &= 0.0016545\end{aligned}$$

$$\therefore R_{eq} = 604.3 \Omega \quad [\text{Matches with table value}]$$

$$I_T = \frac{V}{R_{eq}}$$

$$\text{if } V = 10 \text{ V}$$

$$I_T = \frac{10}{604.3}$$

$$= 16.548 \text{ mA}$$

[Matches theoretical value in table 2.]

$$I_{R_1} = \frac{10}{1000} = 10 \text{ mA}$$

$$I_{R_2} = \frac{10}{2200} = 4.545 \text{ mA}$$

$$I_{R_3} = \frac{10}{5000} = 2 \text{ mA}$$

This matches with theoretical column.

$$I_T = I_{R_1} + I_{R_2} + I_{R_3}$$

$$\Rightarrow 16.7 = 10.14 + 4.66 + 1.97$$

$$\Rightarrow 16.7 \approx 16.77$$

So, KCL is satisfied (almost equal due to practical error).

5. showing all the steps, theoretically calculate R_{eq} . Compare with the experimental value.

Ans: Theoretical value,

$$R_1 = 1 \text{ k}\Omega, R_2 = 2.2 \text{ k}\Omega, R_3 = 5 \text{ k}\Omega$$

$$\begin{aligned} \frac{1}{R_{eq}} &= \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \\ &= \frac{1}{1} + \frac{1}{2.2} + \frac{1}{5} \\ &= 1.6545 \end{aligned}$$

$$\therefore R_{eq} = 0.6043 \text{ k}\Omega$$

$$= 604.3 \Omega$$

experimental value (table: 1)

$$R_{\text{exp}} = 0.604 \text{ k}\Omega$$

$$\therefore 0.604 \text{ k}\Omega \approx 604.3 \Omega$$

so theoretical and experimental values are nearly identical.

6. Calculate all the theoretical values for circuit 2 - Table 4. show all steps.

Ans: From the measurement,

$$V_{\text{dc}} = 15.10 \text{ V}$$

using ohm's law:

$$I = \frac{V}{R}$$

So,

$$R_1 = \frac{V}{I}$$

$$= \frac{7.35 \text{ V}}{7.4 \text{ mA}}$$

$$= 0.993 \text{ k}\Omega$$

$$\approx 1 \text{ k}\Omega$$

$$R_2 = \frac{7.74 \text{ V}}{3.53 \text{ mA}}$$

$$= 2.192 \text{ k}\Omega$$

$$R_3 = \frac{3.77 \text{ V}}{2.84 \text{ mA}} = 1.327 \text{ k}\Omega$$

$$R_4 = \frac{3.95V}{1.85mA}$$

$$= 2.135 \text{ k}\Omega$$

$$R_5 = \frac{20V}{2.83mA}$$

$$= 0.707 \text{ k}\Omega$$

$$R_6 = \frac{2V}{2.82mA}$$

$$= 0.709 \text{ k}\Omega$$

These values are close to standard resistor values considering tolerance.

□ Result Analysis and Discussion:

In this experiment, we constructed two circuits. They were ladder circuits. The first circuit was relatively easier to construct as it consisted of three resistors in parallel and one resistor in series, compared to the second circuit which had a total of six resistors.

For the first circuit, we used ~~four~~ three resistors. Initially, the resistance of the resistors was calculated and recorded using color coding. Then they were placed on the trainer board and their actual resistance was measured using DMM. Circuit 1 was then constructed using the three resistors ($1\text{ k}\Omega$, $2.2\text{ k}\Omega$, $5\text{ k}\Omega$). We had to make sure the wires were connected firmly to the trainer board as abnormal readings were being displayed initially. After successfully constructing the circuit, the DMM was connected along each resistor in series to measure the current flowing across them. We had to make sure the DMM's reading mode was changed to mA, and the red wire was also connected to Ampere. The circuit was then disconnected from the source in order to take the reading for R_{eq} , which is the equivalent resistance of

the circuit.

For table 3, we added the current flowing through each resistor (I_1, I_2, I_3) in parallel and then compared it to that of the resistor in series (I_s). The values were not exactly equal however, the difference was negligible. So we concluded that the sum individual current is equivalent to the total current therefore, verifying the current division rule. We also had to calculate the percentage errors for each reading taken using the formula.

$$\% \text{ Error} = \frac{|\text{Practical value} - \text{Theoretical value}|}{\text{Theoretical value}} \times 100$$

Percentage error is usually caused because of environmental errors, procedure errors, human errors or instrumental errors.

Circuit 2 was then constructed, which look relatively longer than constructing the first circuit. The voltage across each resistor were measured by connecting the DMM parallelly. Then, current across

each resistor was measured by connecting the DMM in series. The values were recorded in table 5.

The experiment was carried out smoothly without any major issues. Minor issues were solved by our lab instructors.

☐ **Conclusion:** Overall, the experiment was successful. By meaning measuring and calculating data from this series parallel circuit and comparing it to the theoretical values both the Kirchhoff's current law (KCL), current divider rule (CDR) using parallel and ladder circuit were verified experimentally.

□ Reference:

1. Fundamental of Electric circuits, Charles K Alexander.
2. Introductory circuit analysis, Robert L. Boylestad.
3. Lab Manual
4. Slides from lab class
5. Data sheet.

□ Attachment:

1. Picture's of the circuits
2. Printed copies of Multisim simulations
3. signed Datasheet.